

Equations

$$v_{ave} = \frac{v_i + v_f}{2}$$

$$v_{ave} = \frac{\Delta x}{\Delta t}$$

$$a_{ave} = \frac{\Delta v}{\Delta t}$$

$$x_f = x_i + v_i t + \frac{1}{2} a t^2$$

$$v_f = v_i + a t$$

$$v_f^2 = v_i^2 + 2a\Delta x$$

$$\vec{r}(T) - \vec{r}(0) = \int_0^{t=T} \vec{v}(t) dt$$

$$\vec{v}(T) - \vec{v}(0) = \int_0^{t=T} \vec{a}(t) dt$$

$$\vec{F}_{net} = m\vec{a}$$

$$\vec{F}_g = m\vec{g}$$

$$\vec{F}_s = -k\vec{x}$$

$$|\vec{F}_{KF}| = \mu_K |\vec{F}_N|$$

$$|\vec{F}_{SF}| \leq \mu_S |\vec{F}_N|$$

$$|\vec{F}_D| \leq \frac{1}{2} C \rho A |\vec{v}|^2$$

$$|\vec{v}| = \sqrt{\frac{2m|\vec{g}|}{C\rho A}} \text{ for vert. fall}$$

$$\vec{p} = m\vec{v}$$

$$\Delta \vec{p} = \vec{F} \Delta t$$

$$\vec{p}_{i,sys} = \vec{p}_{f,sys}$$

$$E_{i,sys} = E_{f,sys}$$

$$V_g = m|\vec{g}|z$$

$$V_s = \frac{1}{2} k x^2$$

$$K = \frac{1}{2} m v^2 + \frac{1}{2} I \omega^2$$

$$W = \vec{F} \cdot \vec{d} = |\vec{F}| |\vec{d}| \cos \theta$$

$$W_{net} = \Delta K$$

$$\theta_f = \theta_i + \omega_i t + \frac{1}{2} \alpha t^2$$

$$\omega_f = \omega_i + \alpha t$$

$$\omega_f^2 = \omega_i^2 + 2\alpha\Delta\theta$$

$$s = r\theta$$

$$|\vec{v}_t| = r|\vec{\omega}|$$

$$|\vec{a}_t| = r|\vec{\alpha}|$$

$$\vec{a} = \frac{d|\vec{v}|}{dt} \hat{v} - \frac{v^2}{r} \hat{r}$$

$$F_c = m a_c = \frac{m v^2}{r}$$

$$|\vec{v}| = \frac{2\pi r}{T}$$

$$I = \sum_i m_i r_i^2$$

$$I = I_{CM} + M d^2$$

$$\vec{L} = I \vec{\omega}$$

$$\vec{F}_g = \frac{G m_1 m_2}{r^2} \hat{r}$$

$$\vec{g} = \frac{GM}{r^2} \hat{r}$$

$$V_g = \frac{G m_1 m_2}{r}$$

$$\vec{F} = -\frac{dV}{dr} \hat{r}$$

$$\vec{F}_E = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}$$

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{r}$$

$$|\vec{E}_{dipole (r \gg d)}| = \frac{1}{4\pi\epsilon_0} \frac{2qd}{r^3}$$

$$V_E = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

$$\vec{F}_E = q \vec{E}$$

Constants --

$$G = 6.67 \times 10^{-11} \frac{Nm^2}{kg^2}$$

$$g = 9.81 \frac{m}{s^2}$$

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \frac{Nm^2}{C^2}$$

$$\epsilon_0 = 8.85 \times 10^{-12} \frac{C^2}{Nm^2}$$

$$c = 3.00 \times 10^8 \frac{m}{s}$$

$$m_{e^-} = 9.11 \times 10^{-31} kg$$

$$m_{p^+} = 1.673 \times 10^{-27} kg$$

$$M_{Earth} = 5.98 \times 10^{24} kg$$

$$R_{Earth} = 6.38 \times 10^6 m$$

$$M_{Sun} = 1.99 \times 10^{30} kg$$

$$R_{Sun} = 6.96 \times 10^8 m$$

$$\rho_{air} = 1.2 \frac{kg}{m^3}$$

Elementary Charge:

$$e = 1.60 \times 10^{-19} C$$

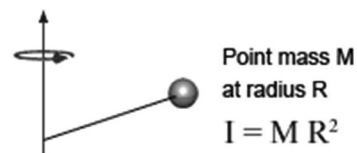
Mathematical Constants:

$$\pi = 3.141592654$$

$$e = 2.718281828$$

Mass Moment of Inertia

Also called *Moment of Inertia*, *Angular Mass*



thin hoop or ring of radius R & mass M:	thick ring of inner radius R1, outer radius R2, and mass M:	solid cylinder or disc of radius R and mass M:	flat plate with sides of length A and B and mass M:
MR^2	$M(r_1^2 + r_2^2) / 2$	$MR^2 / 2$	$M(A^2 + B^2) / 12$
solid sphere of radius R and mass M:	thin-walled hollow sphere of radius R & mass M:	slender rod of length L and mass M, spinning around center:	slender rod of length L and mass M, spinning around end:
$(2/5)MR^2$	$(2/3)MR^2$	$ML^2 / 12$	$ML^2 / 3$